Communication and Internet Technology: From IPv4 to IPv6

IPv6 can be compared with IPv4 in terms of two important parameters, namely, address space and IP header simplification. Internet access is guaranteed with IPv6, even with an exponential growth in Internet users, because there is provision for abundant address space.

Let us begin with a simple question: Why go for IP version 6 (IPv6), when IPv4 is already standardised and is widely used across the globe? The answer is: IPv4 is crippled with challenges. First, it has 32-bit address scheme. Due to an exponential growth of Internet users, IPv4 is running out of address space. The exponential growth rate of Internet users in the world just in the last five years is given below, as per the statistics of Internet World Statistics (IWS)/International Telecommunication Union (ITU):

- In December 2013, total users = 2802 million, percentage of world population = 39
- In December 2014, total users = 3079 million, percentage of world population = 42.4
- In December 2015, total users = 3366 million, percentage of world population = 46.4
- In December 2016, total users = 3696 million, percentage of world population = 49.5
- In December 2017, total users = 4156 million, percentage of world population = 54.4

In IP next generation (IPng) or IP version 6 (IPv6), 128-bit address space (in place of 32 bits of IPv4) and minimum 40-byte IP header (in place of minimum of 20 bytes of IPv4) have been proposed. IPv6 is for coping with increasing demands of Internet access. IP addressing for IPv6 is with 128 bits or 16 bytes.

Second, due to the huge growth of Internet users, organisations are being assigned C addresses in IPv4. This is leading to an explosion in the routing table of the Internet.

Third, IPv4 is for best effort delivery of packet. It neither guarantees packet sequence integrity nor bonded latency in delivery and is, hence, not suitable for real-time services, like voice and video. Thus, with IPv4, quality voice, video and multimedia services are not being serviced.

On the other hand, growth in voice IP traffic is increasingly huge. IPv6 has been proposed to address the stated limitations of IPv4. IPv6 does so with added features, as given below.

- 128-bit address size ensures address space of about 2^{128} compared to theoretical maximum of only 2^{32} in IPv4. IPv6 is believed to support trillions of networks.
- Provision for a single address associated with multiple interfaces.
- Auto-configuration facility.
- Provision for quality of service (QoS) support to real-time services like voice and video, and support to mobility.
- Flow level and priority in the header of IPv6 facilitates support of real-time data.
- Efficient header format (Figure 1) having lesser overhead bits compared to IPv4.
- Fixed header format; header of IPv6 is different from that of IPv4 in many aspects, as follows:

(a) IPv6 has a fixed header size unlike IPv4. Options and padding are variable fields in IPv4. These have been removed in IPv6. This makes IPv6 act in asynchronous transfer mode (ATM) cell in comparison to IPv4, which is a conventional packet in data network.

(b) There is no header checksum in IPv6. This modification is made based on the experience that there is no need for checksum at each intermediate node.